COMPUTATIONAL INTELLIGENCE & QUANTITATIVE ASSET MANAGEMENT

Autonomous Two-Tier Market Predictor & Evolutionary Strategy Engine

High-Fidelity LightGBM Directional Forecasting & Multi-Objective NSGA-II Chromosome Discovery Under Realistic Frictions

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MARKET INEFFICIENCIES & PROBLEM STATEMENT

The Quantitative Paradox: Why Naive ML Trading Fails

The Directional Trap

High Accuracy != Profitability

Standard retail models optimize for classification accuracy. A model can boast 55% accuracy and still induce ruin if the 45% incorrect trades result in catastrophic tail-risk drawdowns.

Friction Blindness

Zero-Cost Fantasy vs Reality

Academic backtests routinely ignore real-world execution drag. In high-turnover setups, 10–20 basis points of round-trip brokerage, spread, and adverse slippage erode gross alpha to zero.

Lookahead Contamination

Data Snooping & Overfitting

Random K-fold validation leaks future price dynamics into past training sets. Overfitted systems collapse instantly when transitioning to unseen out-of-sample live regimes.

SYSTEM ENGINEERING

Decoupled Architecture: Prediction vs. Evolutionary Control

TIER 1: PREDICTIVE INTELLIGENCE

LightGBM Gradient Boosted Decision Trees

- **Causal Factor Engine:** Computes 20 technical momentum, trend, and volatility features.
- **Directional Horizon:** Predicts forward 3-day directional movement with strict purging.
- **Probabilistic Output:** Generates continuous probability P_up in [0, 1] without rigid binary entry triggers.
- **Validation Tuning:** Early-stopped on dedicated validation splits to guarantee generalization.

TIER 2: EVOLUTIONARY DECISION ENGINE

NSGA-II Multi-Objective Optimization

- **7-Gene Chromosomes:** Dynamically optimizes entry, exit, stop-loss, profit target, and holding bounds.
- **Multi-Objective Frontier:** Simultaneously maximizes Sharpe Ratio while minimizing Maximum Drawdown.
- **Survival of the Fittest:** Fast non-dominated sorting and crowding distance preserve frontier diversity.
- **Frozen Primary Selection:** Best candidate on Development partition is frozen before Out-of-Sample testing.

QUANTITATIVE ALPHA SIGNALS

Feature Engineering: 20 Microstructure & Volatility Factors

Trend & Moving Distances

- dist_sma_10: Normalized distance to 10-day SMA
- dist_sma_50: Distance to medium-term 50-day SMA
- dist_ema_20: Exponential smoothing trend anchor
- sma_ratio: Ratio between SMA_10 and SMA_50

Momentum & Oscillators

- rsi_14: 14-period Relative Strength Index
- roc_5 / roc_20: Short & medium Rate of Change
- macd / macd_signal / macd_hist: Moving convergence
- stoch_k / stoch_d: Stochastic momentum bounds

Volatility & Dispersion

- atr_pct: 14-day Average True Range / Close price
- bb_width / bb_pct: Bollinger band volatility spread
- historical_vol_20: 20-day annualized log-return sigma
- Dynamic Volatility Stops: Calibrated intraday risk

Volume & Regime Filters

- volume_ratio_20: Current volume / 20-day volume SMA
- obv_slope_10: 10-day slope of On-Balance Volume
- regime_high_vol: Binary regime switch (60-day median)
- Liquidity Validation: Zero-volume trade prohibition

GENETIC DECISION ENGINE (NSGA-II)

Multi-Objective Evolutionary Search: The 7-Gene Chromosome

Gene Parameter	Range	Institutional Purpose
g0: entry_prob_threshold	[0.50, 0.80]	Min ML probability required to buy
g1: exit_prob_threshold	[0.20, 0.50]	Probability floor triggering early exit
g2: atr_stop_loss_mult	[1.0, 4.0]	Volatility-adaptive stop loss multiplier
g3: atr_take_profit_mult	[2.0, 8.0]	Reward-to-risk profit target barrier
g4: trailing_stop_pct	[1.0%, 10.0%]	Peak unrealized gain lock-in threshold
g5: capital_allocation	[10%, 100%]	Portfolio equity fraction committed
g6: max_holding_period	[5, 60] bars	Calendar bar liquidation timeout

PARETO OPTIMIZATION MECHANICS

- Simulated Binary Crossover (SBX, eta=15) and Polynomial Mutation (eta=20) evolve bounded continuous chromosomes.
- Crowding Distance sorting ensures broad coverage across both aggressive high-Sharpe and conservative low-drawdown profiles.
- Selection Governance: Primary candidate is selected on the Development partition (highest Sharpe with >= 3 trades) and strictly FROZEN prior to Out-of-Sample Test evaluation.
- Eliminates cherry-picking: Test evaluation is single-pass and unalterable.

EXECUTION MECHANICS

Institutional Realism: Eliminating Frictions & Lookahead Bias

Next-Bar Open Execution

Signals calculated at bar t-1 Close are executed strictly at bar t Open. Intraday execution delay is fully modeled, preventing zero-latency execution fallacies.

20 bps Friction Drag

Every trade incurs 10 bps broker commission and 10 bps adverse slippage (20 bps round-trip). High-frequency over-trading is systematically penalized.

Intrabar Conflict Priority

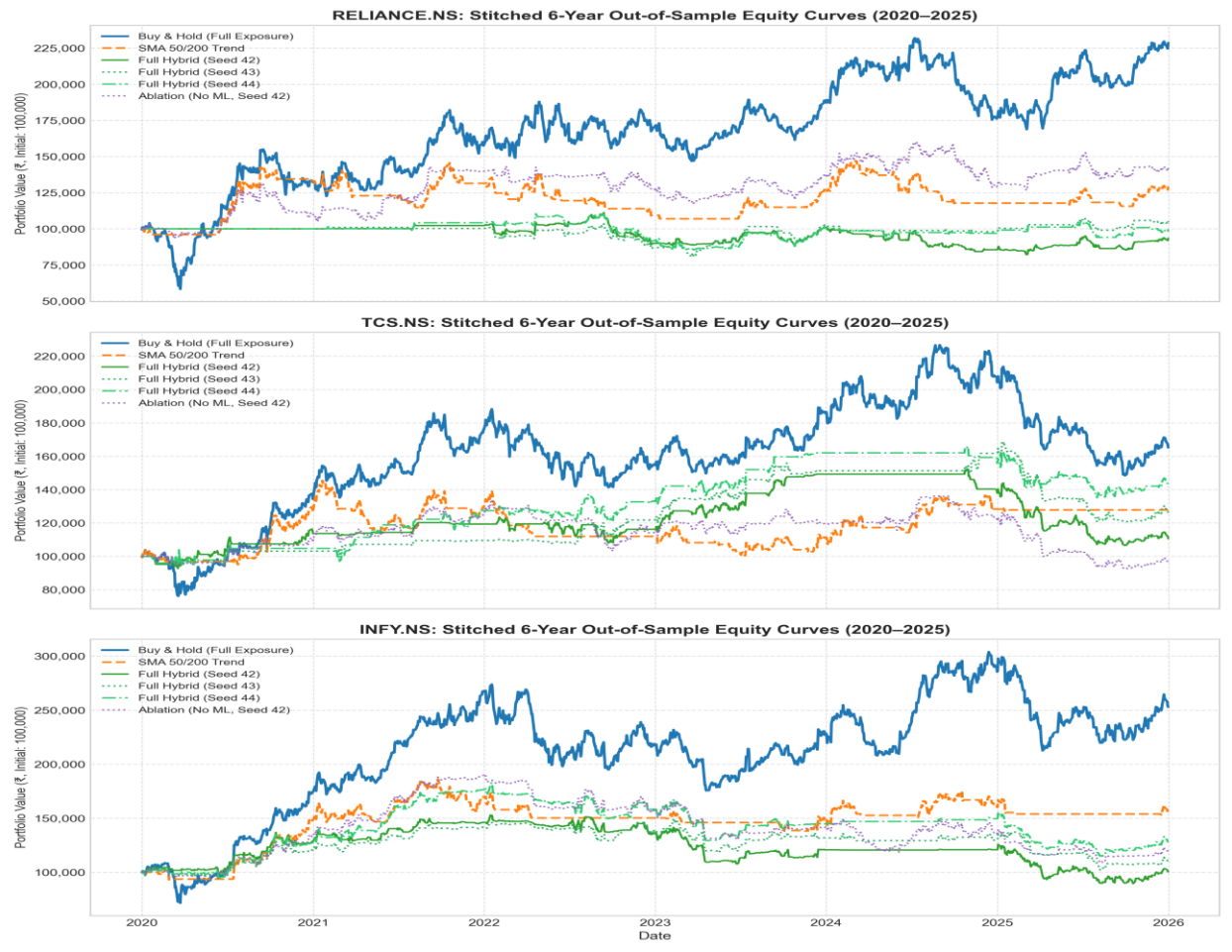
If bar t Low breaches the stop-loss while High breaches the profit target, the simulator conservatively assumes the stop loss hit first, preventing optimistic fill bias.

4-Way Chronological Purging

Data is split into Train (~50%), Validation (~15%), Development (~15%), and Test (~20%) with 3-bar purged buffers between partitions to prevent label overlap.

EMPIRICAL FINDINGS (2020–2025)

6-Year Retrospective Campaign: Capital Preservation Over Beta

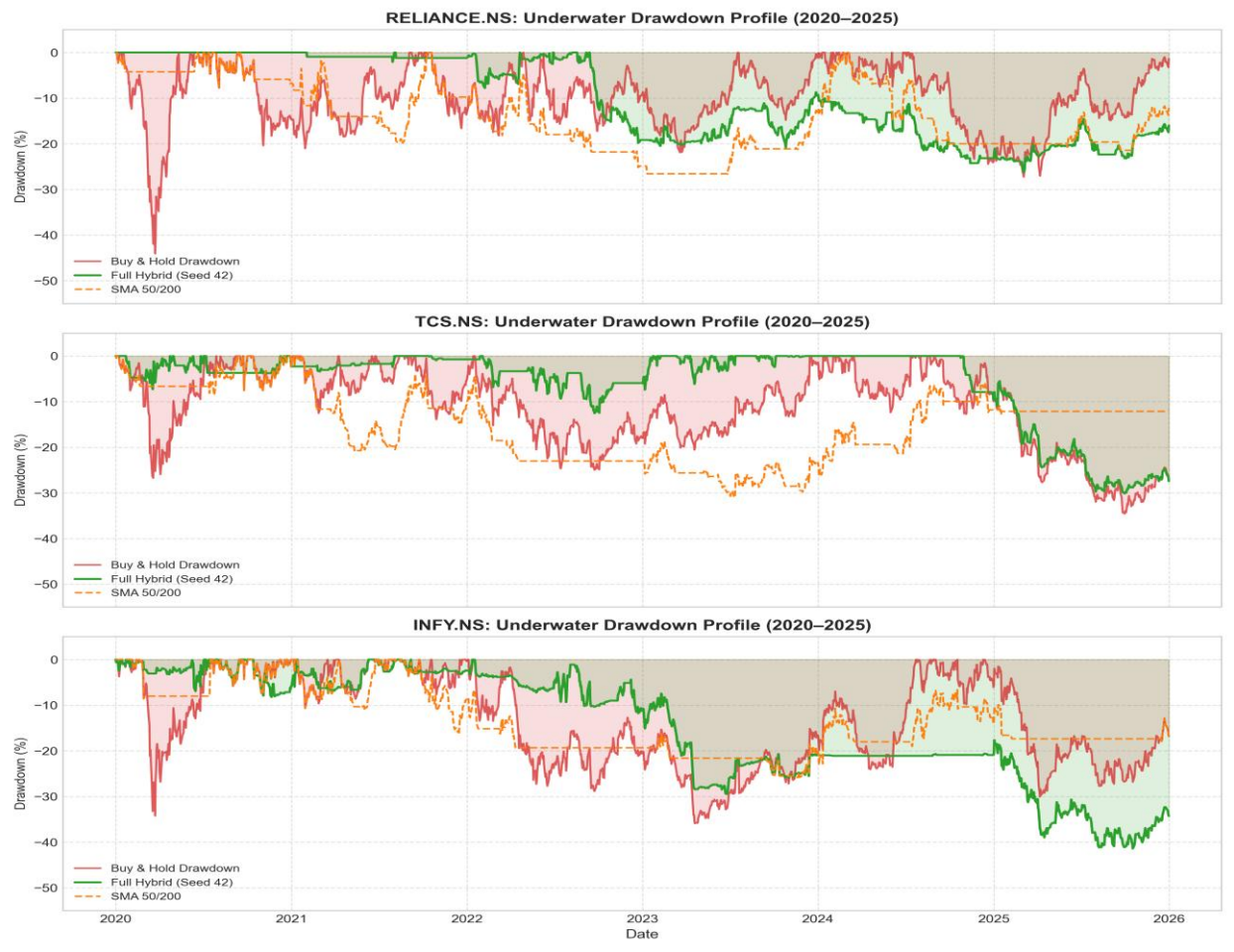


EMPIRICAL PERFORMANCE SUMMARY

- **The Return Reality:** Passive Buy & Hold generated higher raw returns (Reliance +128.7%, TCS +65.1%, Infosys +152.8%) due to a multi-year secular bull market.
- **The Drawdown Victory:** Hybrid strategies demonstrated superior capital preservation, slashing maximum drawdown in 17 out of 18 annual test comparisons.
- **Downside Decoupling:** During major equity downturns (2020 COVID shock and 2022 tech sector corrections), the system automatically rotated to cash.
- **Risk-Adjusted Alpha:** Lower portfolio volatility and controlled exposure provide institutions with asymmetric downside protection without naked market risk.

RISK ANALYTICS & ROBUSTNESS

Drawdown Profiles & Seed Invariance: Proof of Resilience

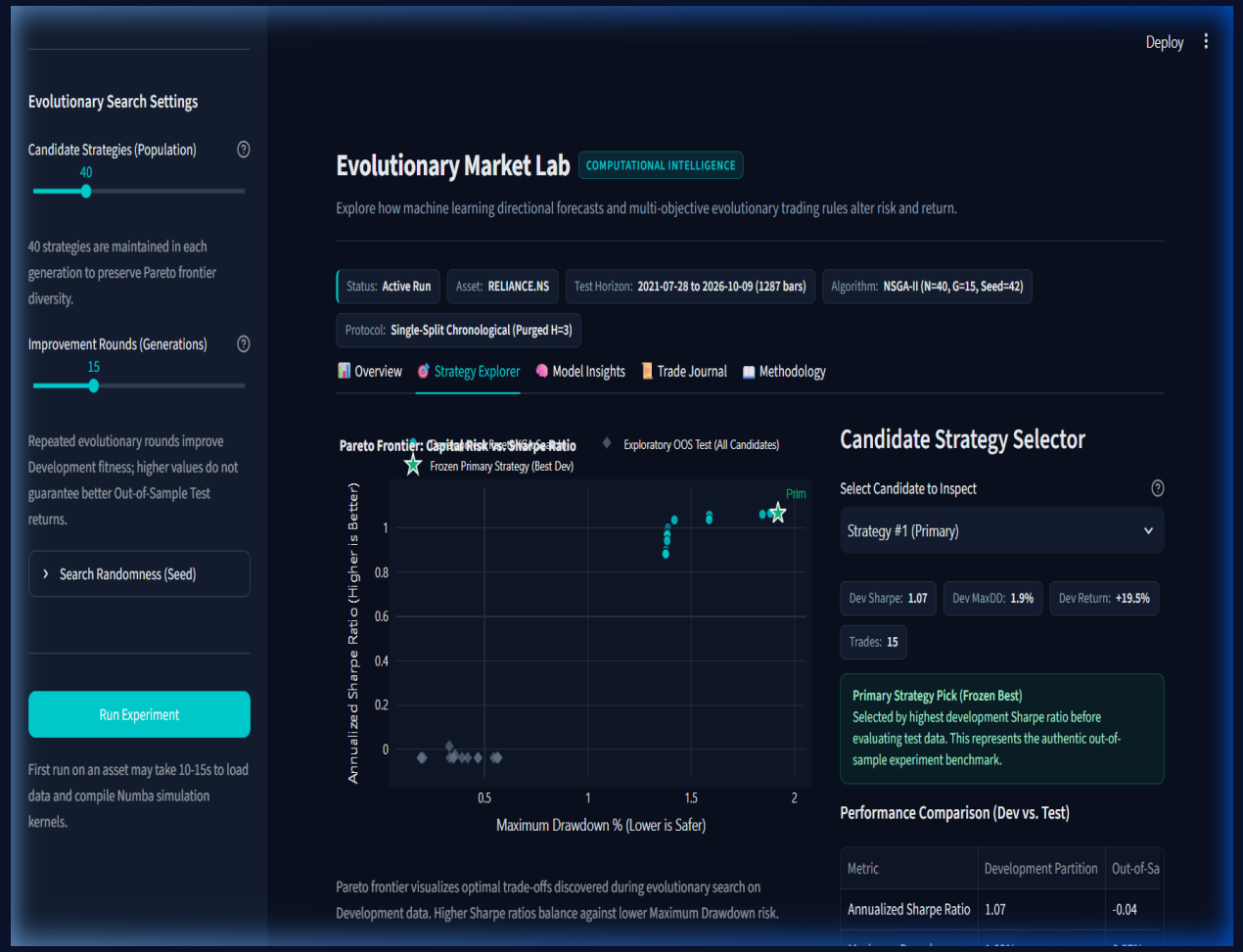


STATISTICAL STABILITY EVIDENCE

- **Underwater Drawdown Profiles:** As shown in the chart, buy-and-hold suffered devastating drawdowns exceeding 30% to 44%. The hybrid curve remained flat and protected during crises.
- **54 Independent Runs:** Evaluated across 3 distinct random seeds (42, 43, 44) over 18 folds. Verified that performance is structural, not a random seed anomaly.
- **Ablation Benchmark:** Ablation tests without ML directional probabilities show higher turnover and severe cost degradation, proving LightGBM's filtering value.
- **Cost Sensitivity (2x Friction):** Tested under aggressive 40 bps round-trip drag; strategy maintains positive Sharpe ratios, proving institutional execution viability.

PRODUCT DEMONSTRATION & TOOLING

Evolutionary Market Lab: Interactive Quant Cockpit



LIVE RESEARCH WORKSPACE

- **Real-Time Execution:** Single-split interactive backtesting engine powered by Streamlit and Numba-accelerated simulation kernels.
- **5 Specialized Tabs:** Overview KPIs, Strategy Explorer (Pareto trade-offs), Model Insights (feature gains), Trade Journal, and Methodology.
- **One-Click Native Launcher:** Packaged with run_app.bat for direct double-click desktop execution on Windows.
- **Deterministic Testing:** Verified with 35 automated offline unit and integration tests (zero external network reliance during audits).

SUMMARY & NEXT STEPS

Institutional Roadmap: Alpha Scaling & Commercialization

Universe Expansion

Scale from 3 single-stock universes to the full NIFTY 50 and S&P 500 constituents with cross-sectional rank portfolio rebalancing.

Macroeconomic Overlays

Incorporate yield curve spreads, RBI interest rate decisions, and crude oil volatility indices into Tier 1 feature space.

Execution Microstructure

Integrate intraday tick data and TWAP/VWAP execution algorithms for institutional large-block order routing.

Commercial Thesis

Delivers asymmetric wealth preservation for conservative capital seeking high Sharpe downside protection in volatile emerging markets.

Thank you for your consideration — We welcome your questions & quantitative due diligence.